Draining Attention or Raising Stress? Brainwave and Cortisol Responses to Varying Phone Placement among NTU Undergraduates

1. Introduction

Many students revise with their phones face down beside their notes. In Singapore, about 30% of residents show problematic smartphone use (Subramaniam et al. 2024), and since January 2026 secondary schools have barred phone use during school hours (MOE 2025). University students, however, face no such restrictions. Ward et al. (2017) proposed that the mere presence of one's own smartphone consumes limited attention, a cost they termed "brain drain", while others suggest that putting the phone away causes stress instead.

2. Literature Review and Research Gap

Evidence for the brain drain effect is mixed. Ward et al. (2017) found lower working memory capacity when participants' phones were placed on their desks than in another room, and Skowronek et al. (2023) reported similar costs for basic attention. However, a replication (Ruiz Pardo and Minda 2022), a recent experiment (Christodoulou and Roussos 2025) and a review of 33 studies (Hartanto et al. 2024) found no significant effect.

The separation account predicts the opposite. In Singapore, Hartanto and Yang (2016) found that phone separation raised anxiety, which impaired executive functions such as attention control. Separation has also produced physical signs of stress in heart activity (Clayton et al. 2015; Konok et al. 2017).

Most of these studies measured task scores, questionnaires or heart activity. Shaer et al. (2024) found that frontal theta waves, a brain signal linked to focused attention, weakened when a smartphone was present, though only in children aged 5 to 8. Cortisol, the body's main stress hormone, appears not to have been measured in any phone separation study. Recording both in the same students would show whether a nearby phone drains attention, a distant phone causes stress, or both.

3. Research Objectives and Questions

This study aims to test both accounts using brain and hormone measures.

RQ1: How do frontal theta activity and reading scores differ when students' phones are on their desk, in their bag or in another room?

RQ2: How does salivary cortisol change across the three placements?

RQ3: To what extent are these differences larger among heavier phone users?

The brain drain account predicts the lowest theta activity and reading scores in the desk group (H1). The separation account predicts the largest rise in cortisol in the other-room group (H2).

4. Scope of the Study

Participants will be full-time NTU undergraduates aged 18 to 25 who own a smartphone. Students taking steroid medication or with hormonal or neurological conditions will be excluded (Riddle et al. 2020). Only short-term effects within a single session will be measured.

5. Methodology

5.1 Design and participants

In this between-subjects experiment, each student is randomly assigned to one placement only, which keeps the conditions from affecting each other. After approval from the NTU Institutional Review Board, 90 students (30 per group) will be recruited through faculty mailing lists and tested over four weeks.

5.2 Measures

A Muse headband, a low-cost device validated for research (Krigolson et al. 2017), will record the brain's electrical activity (EEG). Cortisol in saliva will be measured with a standard laboratory test kit (ELISA). Reading will be scored with two 600-word science

passages, each with five multiple-choice questions. Daily phone use will come from each phone's built-in screen-time record, since self-reports are often inaccurate (Ellis et al. 2019).

5.3 Procedure

Sessions will run between 2 pm and 5 pm, since cortisol varies across the day, and participants will avoid food, caffeine and exercise for an hour beforehand. Participants will be told the study concerns reading and body signals, and the full aim will be explained at the debriefing. After a ten-minute rest, they will give a first saliva sample, and the EEG will record a three-minute baseline. Their silenced phone will then go face down on the desk, into their bag, or to a researcher outside the room. They will read for 20 minutes while the EEG records. A second saliva sample will be taken 25 minutes after the phone was placed, close to when cortisol peaks after a stressful event (Dickerson and Kemeny 2004).

5.4 Data analysis

We will calculate each student's change in frontal theta power from baseline, change in cortisol between samples, and reading score. A one-way ANOVA, which tests whether group averages differ by more than chance, will compare the three groups on each measure, followed by pairwise t-tests. For RQ3, Pearson correlations within each group will test whether heavier phone use is associated with larger theta and cortisol changes.

6. Significance of the Research

To our knowledge, this study will be the first to measure cortisol responses to phone separation and to pair them with brain recordings. If theta activity and reading scores drop in the desk group, students can keep phones off the desk. If cortisol rises mainly when the phone is out of the room, particularly for heavy users, advice to put phones away may need to suit the student. The findings could also guide phone rules in tutorials.

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